ML Foundations — Adult Income Prediction: Problem Framing & Baseline Summary

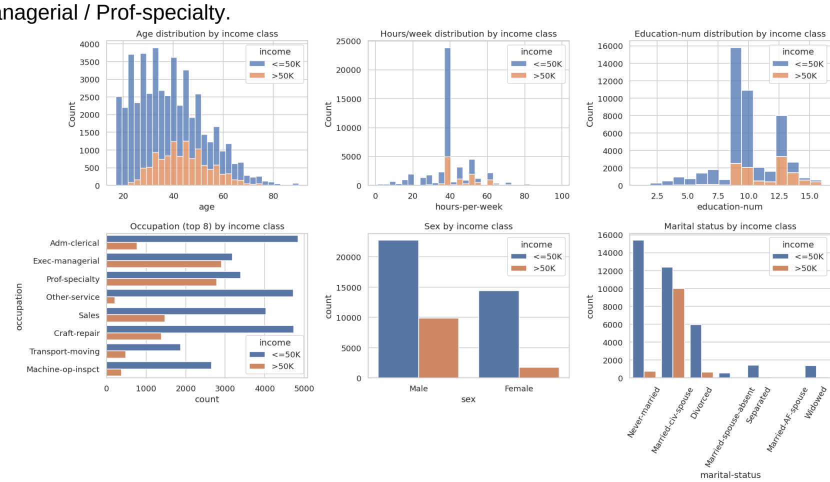
UCI Adult (Census Income) dataset · n = 48,842 · Due 31 Aug 2026

1. Problem Framing & Metric

Target: income >50K/year = positive class (23.9% base rate — imbalanced). **Business objective:** flag likely high earners for a sales/marketing outreach list, where each contact has a real cost, so wasted outreach to the wrong people is expensive. **Primary metric: Precision** (secondary: ROC AUC) — we prioritize correctness of the people we flag over catching every high earner, since false positives directly translate into wasted outreach spend.

2. Data & EDA Highlights

48,842 rows, 14 features. Missing values (as '?') found in **workclass** (5.7%), **occupation** (5.8%), and **native-country** (1.8%) — 7.4% of rows affected overall. Class split: 37,155 (<=50K) vs 11,687 (>50K). Higher income tracks with more years of education, more hours/week, being married, and occupations like Exec-managerial / Prof-specialty.

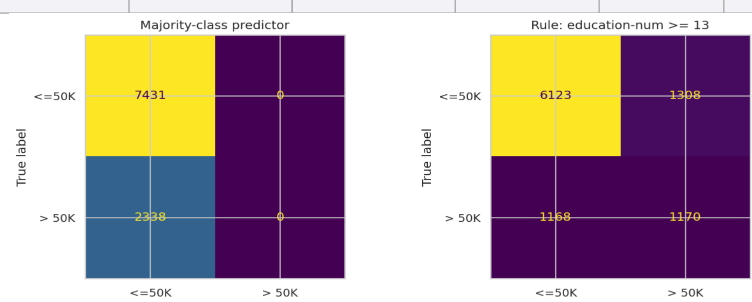


3. Reproducible Split

Stratified 70% train / 10% dev / 20% hold-out test (random_state=42); class rate held at ~23.9% in every split. The hold-out test set is reserved strictly for final evaluation — using it during tuning would let modeling choices implicitly fit that data, inflating the reported score above true generalization performance.

4. Baseline Results (evaluated on hold-out test set, n=9,769)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Majority-class predictor	0.761	0.000	0.000	0.000	0.500	0.239
Rule: education-num >= 13	0.747	0.472	0.500	0.486	0.662	0.356



Interpretation: The majority-class predictor's 76% accuracy is misleading — it has zero precision/recall on the positive class and is business-useless. The single-feature rule (education-num >= 13) trades a little accuracy for real signal: 0.47 precision, 0.66 ROC AUC. A real model should clear roughly **0.60+ precision** (a 10-15 point margin over the rule) without collapsing recall, to justify its added complexity over a one-line rule.

5. Error Analysis & Next Steps

False negatives (1,168, missed by the rule) skew toward married individuals with higher hours/week and 20.5% have capital gains — i.e. self-made/overtime earners without a Bachelor's. **False positives** (1,308) skew younger, lower hours/week, 50% never-married — early-career graduates the rule over-credits.

Fixes for tomorrow: (1) explicit 'Unknown' category for missing workclass/occupation/native-country instead of dropping rows; (2) bucket native-country's ~42 sparse levels; (3) log1p-transform or binarize the zero-inflated capital-gain/loss columns; (4) drop fnlwgt (a census sampling weight, not an individual-level feature); (5) drop redundant education (keep education-num); (6) evaluate class-weighting vs. resampling for the imbalance, tuned on the dev set only.

Primary metric for the week: Precision (+ROC AUC secondary). The rule baseline proves >0.47 precision is achievable from one feature; a real model must push materially past that (target ~0.60+), since precision drives outreach cost — the core business objective. All tuning happens on the dev set; hold-out test is touched once, at the end.